

User Input, If-Else and Switch Case

1. Ask the user for their name and age and print a message like "Hello [Name], you are [Age] years old."
2. Find the maximum of three numbers using nested if-else statements.
3. Calculate an electricity bill. For the first 100 units, the rate is 5 BDT/unit. For units above 100, the rate is 8 BDT/unit. Apply a 10% surcharge on the total bill if consumption exceeds 300 units.
4. Take a student's marks (out of 100) as input and assign a grade using if-else-if:
 - 90-100: A
 - 80-89: B
 - 70-79: C
 - 60-69: D
 - Below 60: F
5. Ask the user for a 4-digit PIN. If the PIN is 1234, print "PIN Accepted." Otherwise, print "Incorrect PIN."
6. Implement a movie ticket pricing system. Standard price is 500 BDT. If the person is a child (age < 12) or a senior citizen (age > 60), give them a 20% discount.
7. Ask for a month number (1-12) and use a switch statement to print the number of days in that month. (Handle February as 28/29 days).
8. Take two numbers and an operator (+, -, *, /) from the user. Use a switch statement to perform the calculation and print the result. [Calculator Using Switch Case]
9. Ask for a username and a password. If the username is "admin" and the password is "password123", print "Login Successful." Otherwise, print "Invalid credentials."
10. Take three side lengths of a triangle as input. Determine if it is equilateral (all sides equal), isosceles (two sides equal), or scalene (no sides equal).
11. Take a number from 1 to 7 as input and use a switch statement to print the corresponding day of the week (e.g., 1 for Sunday, 2 for Monday).
12. Take a character as input and determine if it is an uppercase letter, a lowercase letter, a digit, or a special character.
13. Take the cost price and selling price of a product as input and determine whether there was a profit or a loss.
14. Calculate a person's Body Mass Index (BMI) using their weight (kg) and height (m). Then, classify them based on their BMI using if-else:
 - < 18.5: Underweight
 - 18.5 - 24.9: Normal weight
 - 25 - 29.9: Overweight
 - = 30: Obesity

15. Take a color ("red", "yellow", "green") as input and print the action ("Stop", "Ready", "Go") using if-else-if.

16. If a customer's purchase amount is over 2000 BDT, give them a 15% discount. Otherwise, no discount. Calculate and print the final amount.

17. Take a single digit number (0-9) as input and print it in words using a switch statement.

18. Take three angles of a triangle as input. Check if the triangle is valid (sum of angles must be 180).

19. Find the roots of a quadratic equation $ax^2+bx+c=0$. Consider all cases (real and distinct, real and equal, complex).

20. After a successful PIN entry, give the user options: 1. Check Balance, 2. Withdraw, 3. Deposit. Use a switch case to handle the choice.

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